

Sex emerges at scale

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Attempts to explain the ubiquity of sex in nature always seem to end up in the crowded dustbin of limited-scope theories. We submit that this impasse parallels one that stalled progress on superconductivity for decades. Its resolution required recognition that scale in complex systems is more than a mere vantage point: a parameter that is pivotal at one scale can become asymptotically irrelevant at another. In the evolution of sex, non-additive fitness is such a parameter. It is highly contingent and drives the famously divergent dynamics of sex at sub-scale, yet loses relevance at scale. Once at scale, clarity of hindsight reveals: 1) sex is inevitable, 2) selection for sex follows a universal pattern, and 3) the large costs of sex are paid by the collective behavior of nearly neutral fitness gains.

Evolution of sex | Condensed matter physics | Macro-evolution | Renormalization | Additive fitness

The absence of sex (hereafter *asex*) has several well-studied evolutionary costs that fall naturally into two categories: *within*- and *across*-lineage selective interference. For clarity, we introduce the following notation: the genetic background (or wildtype allele) is denoted $\bullet$ if the lineage is growing and $\circ$ if it has constant size; $+$ denotes advantageous allele, and $-$ denotes deleterious allele. Genotypes have two loci and will be represented by two of these symbols.

Within a growing lineage of genotype $(\bullet, \bullet)$, the efficacy of natural selection is reduced, with the following consequences: 1) decreased fixation probabilities $\mathcal{P}$ of newly-arising beneficial mutants: $\mathcal{P}(\bullet, +) < \mathcal{P}(\circ, +)$, 2) increased fixation probabilities of newly-arising deleterious mutations $\mathcal{P}(\bullet, -) > \mathcal{P}(\circ, -)$, and 3) increased rate $\mathcal{R}$ of irreversible accumulation of deleterious mutations $\mathcal{R}[(\bullet, \bullet) \mapsto (\bullet, -) \mapsto (-, -)] > \mathcal{R}[(\circ, \circ) \mapsto (\circ, -) \mapsto (-, -)]$. Consequences 1 and 2 constitute Hill-Robertson Interference (HRI) after the authors of its first quantitative treatment (1–5); Consequence 3 has been dubbed “lineage contamination” and studied quantitatively in (6) and (7–11).

Across growing lineages, the advantageous traits causing one lineage to grow can be lost if a second lineage appears (not a sublineage of the first) with superior traits. This can happen when: 1) a beneficial mutation arises on a deleterious genetic background, creating competing genotypes $\{(\circ, \circ), (-, +)\}$ or $\{(\bullet, \bullet), (-, +)\}$, 2) one lineage carries a beneficial mutation in one locus and a second lineage carries a beneficial mutation in the other locus, creating competing genotypes: $\{(+, \circ), (\circ, +)\}$ or $\{(+, \bullet), (\bullet, +)\}$. Scenario 1 was envisioned by Fisher, and scenario 2 by Muller; together, they are referred to as Fisher-Muller Interference (FMI). Scenario 2 by itself is often referred to as “clonal interference” (12–17).

Both within- and across-lineage interference are examples of “selection interference” (18) – inefficiencies that arises in the absence of sex. Efficiency in evolution that is lost to selection interference can be regained by breaking the genetic linkage causing it. And linkage breaking is a consequence of sex, defined here as any form of genetic exchange.

The costs of *asex*, while many, are nevertheless thought to be incremental and subtle when compared to the large and obvious costs of sex. Sexually-produced offspring carry only half of the mother’s genes (genome dilution). Sex requires finding a mate in order to reproduce, which makes prospective parents vulnerable to predation and disease. And when only females bear offspring, males are numerically superfluous and impose a canonical two-fold demographic cost (19–23). The cost asymmetry between *asex* and sex constitutes a fitness deficit that clearly favors *asex*.

Significance

Sex is everywhere in nature, despite carrying obvious costs: finding mates, producing males, and passing only half of a parent’s genes to each offspring. Why it persists has puzzled biologists for decades. We show that the answer becomes clearer when evolution is viewed over many populations and long stretches of time. Selection for sex is usually weak, but is occasionally punctuated by very large bursts of advantage. The sizes of these bursts follow a simple power law, making even large costs of sex repeatedly surmountable. This suggests that sex is widespread not because of one special mechanism, but because of a general statistical pattern built into evolution itself.

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The evolutionary advantage of sex is quantified naturally by a dynamical measure proposed by Mustonen and Lässig (24, 25) called “fitness flux”, which can accommodate both fitness landscapes (static topologies) and fitness seascapes (dynamic topologies). We define *sex potential* as the potential fitness flux created by an interference trajectory, or “path” (HRI or FMI). Here we depart from precedent: we concern ourselves not with contingent individual paths but with the collection, or “ensemble”, of all possible paths and the usual probability space induced. Sex potential is therefore a random variable; it is distributed as the *path-ensemble* value of potential fitness flux favoring sex. This quantity reduces to the change in log-odds of a sex^+ modifier frequency.

We consider the selective sorting of n haploid genomes, each of which has two loci, x and y . Here, we depart from precedent by defining genomes as random vectors of genic fitnesses, (X_i, Y_i) , $i = 1, 2, \dots, n$, whose elements need not be independent or identically distributed across loci. (Adding more loci does not change things qualitatively.) We note that this agnostic setting implicitly absorbs all forms of selective interference. Initial genome frequencies can be fixed or random. We derive the average selective advantage of a sex-enabling gene that attaches to a randomly-chosen genome at random time $T \in [0, \infty)$ and subsequently dissociates from that genome with rate $r \in [0, \infty)$.

Mean fitness $L(t)$ may be obtained by partitioning the Fisher-Ewens fundamental theorem of natural selection (FTNS) (26–29) in integrated form as follows:

$$L(t) = T(t) - V(t),$$

where $T(t) = \int_0^t \sigma_X^2(u) + \sigma_Y^2(u) du$ and $V(t) = -2 \int_0^t \sigma_{XY}(u) du$ represent cumulative kinetic- and potential-fitness flux, respectively (24, 25, 30). Under free genetic exchange, $V(t) = 0$ and mean fitness evolves as $L^*(t) = T(t)$. The cumulative fitness flux to be gained from sex at time t is therefore $L^*(t) - L(t) = V(t)$.

Mathematically, *sex potential* is defined as: $V^* := \lim_{t \rightarrow \infty} V(t)$. It reflects total selection for sex-enabling genes that are highly promiscuous, in that they jump from genome to genome through the very recombination they enable. Such genes do not track autosomal (non-sex-related) genes but instead evolve independently in a kind of genetic symbiosis. The fate of sex-enabling genes in this context is therefore determined primarily by population-wide fitness imbalance among autosomal genes.

Not all sex-enabling genes are highly promiscuous, however, with many remaining *linked* for a number of generations to the shuffled autosomal genomes they helped generate. During this time, their frequencies track the autosomal genomes to which they are linked. The result is a process of serial hitchhiking that determines their fate. To probe the full range of promiscuity, we expand our definition of *sex potential* to V_r^* , where r is the dissociation rate of sex-enabling genes. Extremes $r = \infty$ and $r = 0$ correspond to fully promiscuous (no linkage) and fully linked (no promiscuity).

Theorem 1: *Sex potential is universally non-negative in expectation: $\langle V_r^* \rangle \geq 0$.*

The true novelty of the present study lies primarily in the rigorous proof of this theorem (Methods), because it clearly shows why the central claim is non-parametric and does not fall short of universality. There are also two heuristic arguments, however, that cannot get us all the way to universality but that can help to understand why the claim should be true most of the time. The first is a verbal line of reasoning from Otto and Barton (18, 21, 31, 32): they suggest that selectively well-matched alleles ($\sigma_{XY} > 0$) should be rapidly fixed or eliminated, while selectively mismatched alleles ($\sigma_{XY} < 0$) should tend to have longer sojourn times, thereby contributing more to statistical associations. The second heuristic argument is mathematical. Letting random variables ΔX and ΔY denote genic fitness differences at two loci, we have $V^* \propto -\Delta X \Delta Y / |\Delta X + \Delta Y|$. V^* blows up as the denominator approaches zero, which happens as $\Delta Y \rightarrow -\Delta X$, or as $\Delta X \Delta Y \rightarrow -\Delta X^2 \leq 0$. This creates an obvious bias towards positive values of V^* .

Remarkably, sex can initially be strongly deleterious on average ($\langle \sigma_{XY}(0) \rangle \gg 0$) and yet sex potential remains non-negative in expectation ($\langle V_r^* \rangle \geq 0$). This

observation underscores the inadequacy of local measures – such as linkage disequilibrium – for predicting the evolutionary fate of sex at scale.

Theorem 2: *Sex potential is universally heavy-tailed:* $\mathcal{P}(V_r^* > v) \sim v^{-\alpha}$.

Under most realistic conditions, sex potential resides in the universality class $\alpha = 1$. Deviation from this class where $\alpha < 1$ can appear in the (unrealistic) case where, as a fitness peak is approached, the decreasing effect size of beneficial mutations is accompanied by a parallel decrease in effect size of deleterious mutations. In other words, adaptation is necessarily accompanied by an increase in robustness.

Theorem 3: *Sex⁺ correlation length diverges in space and time:*
 $\mathcal{P}(X_r > x) \sim x^{-2}$, and $\mathcal{P}(T_r > t) \sim t^{-1}$.

Taken together, these theorems portray a general pattern governing the evolution of sex at scale: weak but non-negative selection for sex persists in expectation (Thm 1); occasionally, strong surges of selection for sex occur (Thm 2); the strongest of these surges favors linked sex-enforcing genes (1/*r* dependence in Thm 2). Notably, selection surges most likely derive not from large differences at two loci, but from the collective effect of smaller fitness differences across more loci (SI).

The currency of our analyses is additive fitness (27, 33–35). This fact adds to the novelty of the present study by its orthogonality to the non-additive effects, such as epistasis and dominance, often considered essential for sex to be successful on micro-evolutionary scales. Furthermore, additive fitness emerges as the component that survives renormalization and drives the evolution of sex at scale.

To illustrate, we let w_{ij} denote fitness of haploid genome (*i/j*), and we set: $w_{11} = w_{22} = 1$ and $w_{12} = w_{21} = -1$. We consider two extreme scenarios: A) If the population initially consists solely of genotypes (1/1) and (2/2), sex would be detrimental with coefficient of selection -1. B) On the other hand, if the population initially consists solely of genotypes (1/2) and (2/1), sex would be advantageous with coefficient of selection +1. In both cases, there is strong selection on sex genes on micro-evolutionary scales, and yet additive fitness is zero, indicating no net selection at scale. If A were more likely than B, sex would be generally detrimental at scale, and it would indicate that additive fitness does not tell the whole story, i.e., that non-additive fitness is still relevant at scale. The large-scale systemic biases that this would require, however, lack *a priori* rationale (36) and are not supported by the data (37–39). In the language of renormalization, non-additive effects fall naturally into the “irrelevant” bin at scale.

We note that, in the context of organisms that already reproduce sexually, a second interpretation of *ensemble* is possible. Genes or linkage regions in these organisms can evolve independently, and each can experience its own HRI path; a single evolving population of such organisms can thus be considered an ensemble. Sex potential in this context can represent total additive fitness to be gained from increased within-locus recombination in a single population. In this light, it is a measure of selection pressure to maintain sex and potentially reduce gene size (40).

Similar reasoning can be found in the original Hill and Robertson paper (1). The authors present it cautiously, however, because they appear to have doubts about their key assumption which, on the surface, does seem to strain credulity. Namely, they assume that the different HRI paths across loci must have statistical properties that are somehow similar enough to be considered part of the same ensemble. They even invoke rampant gene duplication as a potential scenario that would satisfy their assumption of groupable HRI paths. Had they known that comparable statistical properties are not required, as our results show, they might have been less cautious.

In light of our findings, the observed ubiquity of sex in nature may be interpreted as yet another challenge to assumptions of equilibrium in evolution (41–43). Previous theory that assumes micro-evolutionary equilibrium and a specific kind of fitness mutation finds a “general reduction principle” predicting *ever*-decreasing rates of sex and recombination (44–46). We find

that eliminating the equilibrium requirement at scale reverses the prediction to *never*-decreasing rates of sex and recombination.

When presented with a new theory purporting to explain the evolution and persistence of sex, often the first question that comes to mind is: does it account for and overcome the canonical two-fold cost of sex [REFS]? For many reasons the question itself is problematic and is often dismissed [REFS]. Nevertheless, our answer is clear: the very heavy tail of sex potential means that the two-fold cost (or 3- or 4-fold cost for that matter) is easily overcome repeatedly on evolutionary timescales.

Our theory does not replace existing explanations of sex; it reveals their common large-scale structure. Microevolutionary mechanisms may differ, but the power laws and divergent correlation lengths derived here expose a shared critical organization at scale. A useful analogy is melting: individual molecular trajectories alone do not explain the phase transition. Evolutionary stasis is the frozen state; in the critical regime identified by our power laws, sex^- clusters fragment and sex^+ clusters emerge with scale-free sizes and lifetimes. This boundary regime is therefore especially favourable to the evolution of sex—and perhaps to life itself.

Sex has long seemed exceptional because its causes were sought one mechanism at a time. At scale, those mechanisms resolve into a common law: selection for sex is non-negative on average, punctuated by heavy-tailed surges, and correlated across space and time. The resulting criticality is not assumed but derived, recasting the ubiquity of sex as a universal large-scale pattern rather than the sum of special cases.

Methods

Common notation and kernel reduction. Define $\Delta_X = X_2 - X_1$ and $\Delta_Y = Y_2 - Y_1$, and assume coordinate-reflection symmetry $(\Delta_X, \Delta_Y) \stackrel{d}{=} (\Delta_X, -\Delta_Y)$. Let $q > 0$ denote the initial frequency of the inferior genotype. For any non-negative kernel K , sex potential has the form $V_K = -2q\Delta_X\Delta_Y K(|\Delta_X + \Delta_Y|)$. Reflection of Δ_Y gives $\langle \Delta_X\Delta_Y K(|\Delta_X + \Delta_Y|) \rangle = \langle \mathbf{1}_{\{\Delta_X\Delta_Y > 0\}} \Delta_X\Delta_Y [K(|\Delta_X + \Delta_Y|) - K(|\Delta_X - \Delta_Y|)] \rangle$. On $\{\Delta_X\Delta_Y > 0\}$, $|\Delta_X + \Delta_Y| \geq |\Delta_X - \Delta_Y|$; hence $\langle V_K \rangle \geq 0$ whenever K is non-increasing. For a modifier that remains linked until dissociation at rate r , define $K_{r,t}(s) := (2/s) \int_0^t e^{-ru} (1 + e^{su})^{-1} du$ and $K_r(s) := K_{r,\infty}(s)$. Both kernels are positive and decreasing in s and r . Moreover, as $s \downarrow 0$, $K_{r,t}(s) \sim k_{r,t}/s$, where $k_{r,t} := \int_0^t e^{-ru} du = (1 - e^{-rt})/r$, with $k_{0,t} = t$; thus, for $r > 0$, $K_r(s) \sim k_r/s$ with $k_r = 1/r$. For Theorems 2 and 3, set $U = \Delta_X + \Delta_Y$, $W = \Delta_X$, and $S = |U|$. Assume that (U, W) has a joint density $f_{U,W}$ continuous near $U = 0$, that $M := \int_{\mathbb{R}} w^2 f_{U,W}(0, w) dw \in (0, \infty)$, and that S has a density g continuous at zero with $g(0) \in (0, \infty)$.

Proof of Theorem 1. For the promiscuous modifier, $K(s) = 1/s$ is decreasing; for dissociation rate r , the integral representation above shows that $K_r(s)$ is decreasing. The kernel reduction therefore gives $\langle V(\infty) \rangle \geq 0$ and $\langle V_r(\infty) \rangle \geq 0$ for every r .

Proof of Theorem 2. Since $-\Delta_X\Delta_Y = W^2 - UW$, the finite-time potential is $V_r(t) = 2q(W^2 - UW)K_{r,t}(|U|)$. As $U \rightarrow 0$, $V_r(t) \sim 2qk_{r,t}W^2/|U|$, so $V_r(t) > v$ is asymptotically equivalent to $|U| < 2qk_{r,t}W^2/v$. Continuity of $f_{U,W}$ at $U = 0$ therefore gives $\mathbb{P}[V_r(t) > v] \sim c_{r,t}v^{-1}$, where $c_{r,t} := 4qk_{r,t}M$. Thus $\alpha = 1$ and, for $r > 0$, $c_{r,t} \rightarrow c_r = 4qM/r$ as $t \rightarrow \infty$.

Supplementary materials

Materials and Methods

Supplementary Text

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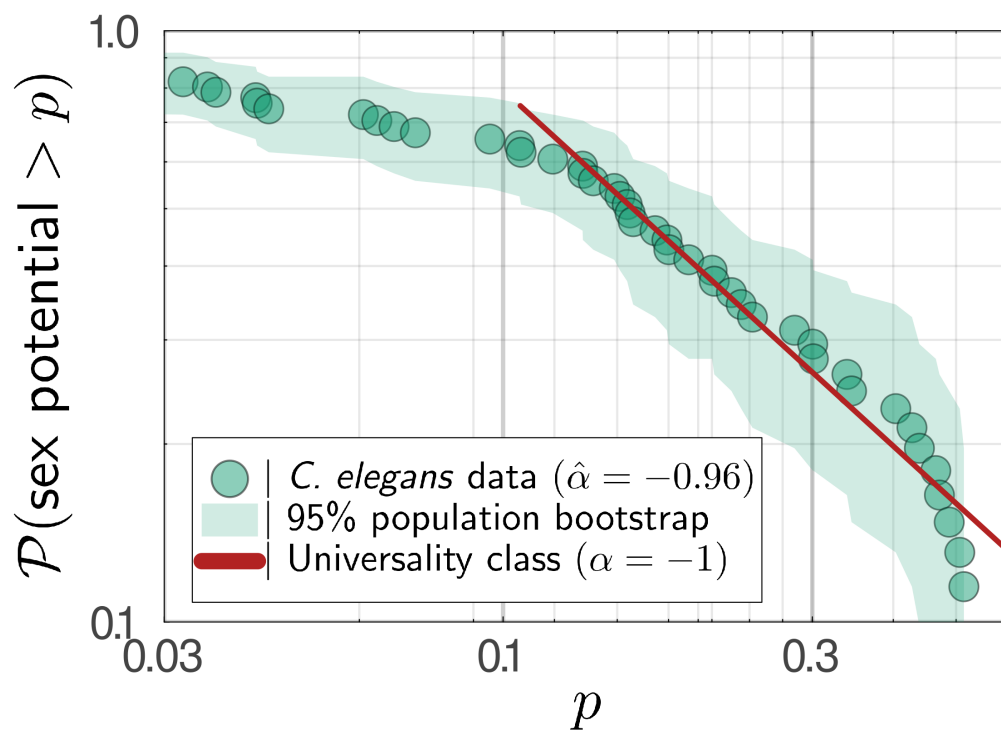


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